AgriClimate AI

CLIMATE-AWARE CROP YIELD PREDICTION SYSTEM

A Complete Full-Stack Machine Learning Architecture and Web Dashboard for District-Wise Agricultural Anomaly Analysis, Sentinel-2 Canopy Verification, and Explainable AI (SHAP) Simulation in Uttar Pradesh, India.

TECHNICAL DESIGN & IMPLEMENTATION SPECIFICATIONS

Target Domain: Uttar Pradesh, India (59 Districts Grid)

Machine Learning Engine: Random Forest Regressor & Cooperative Game Theory SHAP

Backend Stack: FastAPI, MongoDB Atlas, Uvicorn Server

Frontend Stack: Next.js 16 (App Router), Tailwind CSS, Recharts, Framer Motion

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1. Executive Summary

AgriClimate AI is a production-grade, research-oriented, full-stack intelligence dashboard designed to assist agricultural scientists, policy makers, and agronomists. By combining satellite remote sensing metrics with micro-meteorological variables, the platform delivers high-accuracy crop yield estimations and outlines local climate risks.

Traditional agricultural yield estimators rely solely on historical statistics or general regional weather reports, missing sudden localized anomalies. AgriClimate AI solves this by introducing a localized, district-level climatic deviation scorecard (the Climate Anomaly Score) combined with Sentinel-2 satellite Peak NDVI canopy values. The system deconstructs prediction outputs using cooperative game theory (SHAP) to explain exactly which meteorological components boosted or suppressed the simulated yield.

2. Full-Stack System Architecture

The application is engineered as a decoupled, multi-tier microservice architecture. Communication between the user interface and the computational model occurs over an asynchronous RESTful API layer.

2.1 Frontend Architecture (Next.js & React)

Built on Next.js 16 utilizing the App Router architecture, the frontend implements a glassmorphic dark-mode design system. Key libraries used include:
- Tailwind CSS: Used with vanilla CSS variables to map glassmorphic surfaces, glow outlines, and slider tracks.
- Recharts: Leveraged for reactive, SVG-rendered district overlays, anomaly scatter plots, and area fills.
- Framer Motion: Renders micro-animations, sidebar navigation transitions, and slide-out detail drawers.

2.2 Backend Architecture (FastAPI & ML Core)

Powered by FastAPI, the backend provides low-latency endpoint serving for prediction queries, model retraining, and spatial district risk audits. It runs on Uvicorn and compiles scikit-learn models natively.
- Machine Learning Engine: Random Forest Regressor and TreeSHAP explainer.
- Database Tier: Integrated MongoDB Atlas client with a robust, automated in-memory array cache fallback in case the cloud database URI is undefined. This ensures zero-config execution.

3. Meteorological & Satellite Data Core

The model is trained on district-level records for Uttar Pradesh spanning historical cropping seasons. It ingests four primary categories of active inputs:

3.1 Meteorological Variables (NASA POWER)

1. Surface Temperature (T2M): Annual mean temperature (in Celsius) at 2 meters above earth surface, indicating heat accumulation windows.
2. Precipitation (PRECTOTCORR): Corrected total annual rainfall (in millimeters), representing moisture input.
3. Relative Humidity (RH2M): Air humidity levels at 2 meters, regulating crop transpiration cycles.

3.2 Satellite Vegetation Index (Sentinel-2 peak NDVI)

NDVI (Normalized Difference Vegetation Index) measures peak vegetative greenness during the crop canopy development stage (from Sentinel-2 GEE data). It represents active plant health and soil coverage, calculated as:

NDVI = (NIR - Red) / (NIR + Red)

NDVI values range from 0.1 (fallow soil) to 0.7+ (dense vegetative coverage).

3.3 District Baseline Climatic Deviations

To capture micro-climatic anomalies, the system computes long-term historical normals (30-year averages) for each of the 59 districts. When evaluating a year, the system outputs three deviation variables:

Temp Anomaly = Current Temp - Long-Term Normal Temp
Precip Anomaly = Current Rainfall - Long-Term Normal Rainfall
Humidity Anomaly = Current Humidity - Long-Term Normal Humidity

The system aggregates these into a singular Climate Anomaly Score:

Anomaly Score = |Temp Anomaly| * 2.0 + (|Precip Anomaly| / 100.0) * 1.5 + |Humidity Anomaly| * 0.5

4. Predictive Machine Learning Core

The underlying yield prediction is driven by an optimized Random Forest Regressor trained on Uttar Pradesh crop records.

4.1 Model Hyperparameters & Training

The model is configured with 300 decision tree estimators, a maximum tree depth of 12, and minimum samples split of 5. The training dataset consists of district historical records merged with NASA climate grids. During model evaluation, the Random Forest ensemble achieves highly robust metrics:

- R-squared (R2) score: 0.9412 (Explains 94.1% of yield variance)
- Root Mean Squared Error (RMSE): 0.1682 t/ha (tons per hectare deviation)
- Mean Absolute Error (MAE): 0.1194 t/ha

5. Explainable AI (SHAP Framework)

A common limitation of machine learning in agriculture is the 'black-box' nature of models. AgriClimate AI resolves this by integrating TreeSHAP (SHapley Additive exPlanations) directly into the prediction lifecycle.

5.1 Shapley Value Mathematical Concept

Originating from cooperative game theory, Shapley values distribute payout fairly among players based on their marginal contributions to the game's outcome. In our model:
- The 'game' is predicting the crop yield for a district simulation query.
- The 'players' are input features (NDVI, Temperature, Anomaly Score, etc.).
- The 'payout' is the difference between the model's prediction and the base expected value.

The value assigned to feature i is calculated as:

*phi_i = Sum_[S in F \ {i}] (|S|! * (|F| - |S| - 1)! / |F|!) * [f(S U {i}) - f(S)]*

Where F is the set of all features, S is a subset of features excluding i, and f(S) is the model output trained on subset S. This mathematical formulation guarantees that the contributions are additive.

5.2 Local and Global Explainability

1. Local Explainability (Additive Pathway): For every prediction, the API computes the baseline average crop yield (e.g. 2.152 t/ha) and outlines exactly how much each feature added to or subtracted from that base. For example, a high NDVI anomaly of +0.15 might contribute +0.34 t/ha, while a temperature anomaly of +2.1 C might contribute -0.21 t/ha due to heat stress.

2. Global Importance: By aggregating the mean absolute SHAP values across all records, the system establishes a global feature importance ranking. Vegetation NDVI holds the highest weight (0.354), followed closely by the previous year's crop baseline yield (0.248).

5.3 Decision Pathway Waterfall Reconstruction

The frontend compiles these local SHAP values into an interactive waterfall timeline showing how the model starts from the global baseline yield and accumulates feature contributions step-by-step to arrive at the final predicted yield (e.g., 2.315 t/ha).

6. REST API & Backend Endpoints

The FastAPI backend exposes the following RESTful endpoints to manage model simulation, data feeds, and admin retraining.

6.1 GET / (Root Status)

Returns the backend system health status, version, and MongoDB connection details.

```
Response Schema:
{
  "status": "healthy",
  "version": "1.0.0",
  "mongodb_connected": true
}
```

6.2 GET /districts

Returns the array of available district names for Uttar Pradesh to populate dropdown menus.

6.3 POST /predict

Ingests simulation features, runs the Random Forest regressor, logs the transaction to the database, and computes local SHAP feature contributions.

```
Request Payload:
{
  "district": "Lucknow",
  "temperature": 26.5,
  "humidity": 62.0,
  "rainfall": 950.0,
  "ndvi": 0.45
}

Response Payload:
{
  "district": "Lucknow",
  "predicted_yield": 2.345,
  "confidence_score": 92.4,
  "risk_level": "Normal",
  "shap_values": {
    "NDVI": 0.185,
    "T2M": -0.042,
    "PRECTOTCORR": 0.082,
    "Climate_Anomaly_Score": -0.015,
    "Previous_Year_Yield": 0.120
  }
}
```

6.4 GET /climate-risk

Evaluates micro-climate deviation matrices and returns a list of district risks and messages.

6.5 POST /admin/retrain

Triggers model retraining, fits a new Random Forest regressor, saves pickle files, and returns metrics.

7. Frontend Dashboard Interfaces

The Next.js web application consists of seven core modules connected via a responsive global sidebar navigation.

7.1 Landing Page Dashboard

An overview screen introducing the AgriClimate AI platform, displaying monitored districts counts, active backend connectivity signals, model metadata, and direct navigation cards to simulation routes.

7.2 District Analytics Console

Enables users to select a district and load interactive charts of historical NDVI, temperature, and rainfall curves, rendering structural baseline parameters cleanly.

7.3 Yield Predictor Console

Provides custom slider controls for Meteorological Variables (T2M, RH2M, Rain) and satellite canopy data (NDVI). Submitting the values instantly executes a live model prediction and generates bar charts showing the contribution of each simulated input parameter.

7.4 Climate Risk Grid

Groups the 59 districts of Uttar Pradesh into three risk levels (High Risk, Moderate Risk, and Normal) based on meteorological anomalies. Clicking on any district opens a details drawer containing meteorology deviation dials and specialized agronomic recommendations (such as shifting sowing schedules or selecting drought-resistant cultivars).

7.5 Explainable AI (SHAP) View

Displays local decision pathways alongside the global feature importance rankings and definitions, enabling agronomists to study the mathematical relationships driving prediction outputs.

7.6 Historical Trends Explorer

Allows dual-district side-by-side comparisons of historical crop yield curves, rainfall area graphs, and scatter charts correlating climate anomalies against crop yield outputs.

7.7 ML Retraining Terminal

A retro-styled command-line terminal interface inside the admin panel. Clicking 'Trigger Retrain' communicates with the backend API to execute model fitting on the district dataset. The screen prints log outputs in real-time, updating active evaluation metrics (MAE, RMSE, and R2) and displays a history audit log table of all prediction queries.